Calculus na 26.10.2026

Zadanie	1	2	3	4	5	6	7	8	9	10	Suma
Punkty											

Zadanie 1:

Compute the second derivatives in Exercises 27–32.

29. $\dfrac{d^2}{dx^2}\,\dfrac{x^2+1}{x+2}$

Zadanie 2:

a)

Use the chain rule rule to differentiate the functions in Exercises 25–34.

25. $(x^2 - 6x + 1)^3$

c)

27. $\dfrac{9 + 2x^5}{3 + 5x^5}$

Zadanie 3:

Use the chain rule rule to differentiate the functions in Exercises 25–34.

33. $\sqrt{4x^5 + 5x^2}$

Zadanie 4:

a)

★52. Prove that $(d/dx)(u^n) = nu^{n-1}du/dx$ for all natural numbers n as follows:

(a) Note that this is established for $n = 1, 2, 3$ at the beginning of this section.

b)

(b) Assume that the result is true for $n - 1$, and write $u^n = u(u^{n-1})$. Now differentiate using the product rule to establish the result for n.

c)

(c) Use induction to conclude the result for all n. (See Exercise 65, p. 69.)

Zadanie 5:

53. Find a general formula for $(d^2/dx^2)(u^n)$, where $u = f(x)$ is any function of x.

Zadanie 6:

55. Let f and g be functions such that $f \circ g = i$, where i is the function in Problem 54. Find a formula for $f'(x)$ in terms of the derivative of g.

Zadanie 7:

36. If $x^3 + y^3 = xy$, compute dx/dy in terms of x and y.

Zadanie 8:

a)

37. Suppose that $x^4 + y^2 + y - 3 = 0$.

(a) Compute dy/dx by implicit differentiation.

b)

(b) What is dy/dx when $x = 1, y = 1$?

c)

(c) Solve for y in terms of x (by the quadratic formula) and compute dy/dx directly. Compare with your answer in part (a).

Zadanie 9:

41. Find the equation of the tangent line to the curve $x^4 + y^4 = 2$ when $x = y = 1$.

Zadanie 10:

★60. Using implicit differentiation, find the equation of the tangent line at the point (x_0, y_0) on the circle $(x - a)^2 + (y - b)^2 = r^2$. Interpret your result geometrically. (a, b, and r are constants.)